

Adaptive Krylov–Lanczos Exponential Integration for Sparse QFLPN State Evolution: Long-Horizon Accuracy, Step Control, and Work Efficiency

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Abstract

Long-horizon propagation of a sparse state requires a numerical strategy that controls both the approximation dimension and the time-step length. A fixed Krylov dimension can be unnecessarily expensive when the local exponential action is easy, whereas a fixed time step can over-resolve some intervals and under-resolve others. This paper develops and experimentally evaluates an adaptive exponential integrator based on symmetric Lanczos projection and step-doubling error control. The numerical problem is the matrix-free evolution $y'(t) = Ay(t)$ with A a sparse symmetric negative-semidefinite operator compatible with a linearized QFLPN transition layer. The method constructs a small tridiagonal Lanczos projection, evaluates the exponential of that projected matrix, and compares one full step with two half steps to obtain a computable local discrepancy indicator. The experiment uses a deterministic normalized initial state, dimension $N = 3,000$, final time $T = 20$, and three operator scales. Fixed-step Lanczos and Arnoldi configurations are compared with adaptive Lanczos propagation. For scale $s = 4$, adaptive propagation with Krylov dimension $m = 28$ accepted four steps and rejected none, achieving a relative final error of 2.43×10^{-14} with 336 sparse operator applications. At $m = 20$, the same adaptive strategy accepted seven steps and rejected three, with final relative error 7.17×10^{-14} and 420 operator applications. The experiments show that symmetry can be exploited to replace a general Hessenberg projection by a tridiagonal one, while step adaptation reduces the number of global propagation stages. The principal contribution is a reproducible accuracy–work protocol that separates local step control, Krylov dimension, and global error verification rather than treating them as a single tuning parameter.

Keywords: QFLPN; Krylov subspace; Lanczos method; exponential integrator; adaptive time stepping; sparse operators; matrix exponential; long-horizon propagation; reproducibility

1 Introduction

A sparse state-evolution model may require the repeated action of a matrix exponential rather than a single evaluation of $e^{tA}v$. The distinction matters when the propagation horizon is long, when the operator scale changes the local difficulty, or when the desired accuracy must be maintained over many successive steps. In such situations, two numerical parameters interact: the dimension of the approximation space and the length of the propagation step.

Krylov methods provide a natural matrix-free framework for this task. Saad established the projection approach for matrix-exponential actions together with a priori and a posteriori error analysis, while Hochbruck and Lubich developed convergence results for both Arnoldi and Lanczos approximations and connected the approach with exponential time integration. [1, 2] Later adaptive Krylov exponential-integrator algorithms made the coupling between time-step size and Krylov dimension explicit. [3, 4]

The present work addresses a narrower and experimentally testable question: for a symmetric sparse transition operator, how should Krylov dimension and time-step length be coordinated over a long propagation interval? The study focuses on four quantities that can be measured independently: final relative error, accepted and rejected time steps, sparse operator applications, and wall-clock time.

The paper has four scientific objectives. First, it formulates a symmetric Lanczos exponential step using only sparse matrix–vector products. Second, it introduces a step-doubling controller that compares one step of length h with two steps of length $h/2$. Third, it compares adaptive propagation with fixed-step Lanczos and general Arnoldi propagation under the same operator family and final-time reference. Fourth, it identifies the regime in which the additional cost of adaptivity is justified by reduced global work.

The scientific object is the numerical integration layer. The QFLPN semantic layer is treated as the source of a sparse linear operator; no claim is made here about the formal semantics, quantum advantage, physical hardware, or universal superiority of one matrix-function algorithm.

2 Problem Formulation and Scientific Boundary

Consider the autonomous linear system

$$\frac{dy(t)}{dt} = Ay(t), \quad y(0) = v, \quad (1)$$

with exact solution

$$y(T) = e^{TA}v. \quad (2)$$

The operator is assumed sparse and symmetric,

$$A = A^T, \quad (3)$$

so that a three-term Lanczos recurrence is available. The primary family is

$$A_s = s \text{ tridiag}(1, -2, 1), \quad (4)$$

with $N = 3,000$, $s \in \{0.5, 2, 4\}$, and $T = 20$.

The state dimension N is a numerical representation dimension. It is not interpreted as a qubit count. This distinction is necessary because a QFLPN may generate a sparse numerical operator whose dimension is determined by the chosen representation rather than by a direct dense quantum-state encoding.

The operator is negative semidefinite. Consequently, the exact evolution is contractive in the Euclidean norm for this benchmark family. This makes the experiment useful for isolating numerical approximation effects without introducing growth caused by an unstable continuous-time operator.

The paper deliberately does not reuse the million-element CSR/SpMV timing experiment as its scientific result. Sparse storage and single-step matrix–vector throughput are not the research question here. The present result concerns long-horizon matrix-function integration and the interaction between Krylov dimension and time-step adaptation.

3 Spectral Structure of the Benchmark Operator

The chosen tridiagonal family has an analytically known spectrum. For the Dirichlet second-difference matrix,

$$\lambda_k(A_s) = -4s \sin^2\left(\frac{k\pi}{2(N+1)}\right), \quad k = 1, \dots, N. \quad (5)$$

Hence

$$-4s < \lambda_k(A_s) < 0, \quad (6)$$

with the largest magnitude eigenvalue approaching $4s$ as N grows. This gives an exact spectral scale for the controlled experiment and explains why increasing s makes the exponential action numerically more demanding at fixed T .

Theorem 1 (Contractivity of the reference family). *For $A_s = s \operatorname{tridiag}(1, -2, 1)$ with $s > 0$, the semigroup satisfies*

$$\|e^{tA_s}\|_2 \leq 1, \quad t \geq 0. \quad (7)$$

Proof. The matrix A_s is real symmetric with non-positive eigenvalues. Therefore it is orthogonally diagonalizable as $A_s = U\Lambda U^\top$, and $e^{tA_s} = Ue^{t\Lambda}U^\top$. Every diagonal entry of $e^{t\Lambda}$ lies in $(0, 1]$, so its spectral norm is at most one. $\square$

The contractivity property is important for interpretation. It prevents the exact dynamics from amplifying the state norm and makes the final reference error a direct measure of numerical propagation quality rather than a mixture of numerical error and physical instability.

Table 1: Analytical spectral range of the benchmark family.

s	Approximate spectral interval	$\ A_s\ _2$
0.5	$(-2, 0)$	≈ 2
2	$(-8, 0)$	≈ 8
4	$(-16, 0)$	≈ 16

4 Lanczos Exponential Step

For a nonzero state y , define

$$\beta = \|y\|_2, \quad q_1 = \frac{y}{\beta}. \quad (8)$$

The Lanczos recurrence generates orthonormal vectors q_j and coefficients α_j, β_j such that

$$AQ_m = Q_m T_m + \beta_m q_{m+1} e_m^\top, \quad (9)$$

where T_m is symmetric tridiagonal,

$$T_m = \begin{bmatrix} \alpha_1 & \beta_1 & & & \\ \beta_1 & \alpha_2 & & & \\ & & \ddots & & \\ & & & \ddots & \beta_{m-1} \\ & & & \beta_{m-1} & \alpha_m \end{bmatrix}. \quad (10)$$

The projected exponential action is

$$\Phi_m(h, y) = \beta Q_m e^{hT_m} e_1. \quad (11)$$

Only the small dense exponential e^{hT_m} is formed. The large matrix e^{hA} is never constructed.

The implementation uses modified orthogonalization with a second correction pass. The second pass is inexpensive for the present moderate dimensions and protects the orthogonality of the basis from accumulated floating-point loss. The projected matrix remains symmetric to numerical precision.

Proposition 1 (Projection consistency). *If the Lanczos basis is orthonormal and the recurrence is exact in arithmetic, the first m projected moments of the exponential action are represented by the tridiagonal matrix T_m through equation (11).*

Proof. The Lanczos construction gives $Q_m^\top A Q_m = T_m$. Therefore every polynomial p of degree at most $m - 1$ satisfies $Q_m^\top p(A) q_1 = p(T_m) e_1$. The exponential is approximated by its polynomial action in the projected space, yielding the stated representation. $\square$

5 Adaptive Exponential Integration

A fixed Krylov dimension controls the approximation space but does not control the global propagation error when many steps are composed. We therefore use an embedded step-doubling construction.

Given a current state y_n and step h , compute

$$y^{(1)} = \Phi_m(h, y_n), \quad (12)$$

and independently compute two half steps,

$$y^{(1/2)} = \Phi_m(h/2, y_n), \quad y^{(2/2)} = \Phi_m(h/2, y^{(1/2)}). \quad (13)$$

The local discrepancy indicator is

$$\eta_n = \frac{\|y^{(2/2)} - y^{(1)}\|_2}{\max(\|y^{(2/2)}\|_2, 10^{-300})}. \quad (14)$$

The two-half-step result is accepted when $\eta_n \leq \tau_{\text{loc}}$. Otherwise the step is rejected and recomputed with a smaller h .

The controller updates the step using

$$h_{\text{new}} = h \text{ clip} \left(0.9 \left(\frac{\tau_{\text{loc}}}{\eta_n} \right)^{1/3}, 0.25, 2.0 \right), \quad (15)$$

with explicit lower and upper bounds. The exponent $1/3$ is a practical controller choice for the discrepancy between a full step and two half steps; it is not presented as a sharp global error theorem for the projected method.

Definition 1 (Accuracy-constrained adaptive propagation). *For a prescribed local tolerance τ_{loc} , the adaptive method accepts the two-half-step solution when equation (14) satisfies the tolerance and continues until $t = T$. The final global error is evaluated independently against a high-accuracy reference and is not inferred solely from the local indicator.*

This separation is important. The local indicator controls the algorithm, whereas the final reference comparison validates the resulting global trajectory. The two quantities answer different questions and should not be conflated.

6 Work Model

For one Lanczos step with dimension m , the dominant large-vector operation is one sparse matrix-vector product per generated basis vector. A propagation with n accepted steps and no rejected work therefore has approximately

$$C_{\text{Lanczos}} \approx nm \quad (16)$$

operator applications. The adaptive step-doubling controller performs additional trial work, so rejected steps and the two-half-step construction must be counted explicitly.

For a fixed-step Arnoldi implementation with the same dimension, the number of sparse operator applications is also approximately nm . The difference is in the basis construction: Arnoldi stores a general upper-Hessenberg projection and performs a growing set of orthogonalization operations, whereas Lanczos exploits symmetry and produces a tridiagonal projection.

We therefore report three distinct cost measures:

$$C_{\text{op}} = \text{number of sparse operator applications}, \quad (17)$$

$$C_{\text{trial}} = \text{number of accepted and rejected adaptive trials}, \quad (18)$$

and observed wall-clock time. No one of these quantities is treated as a universal performance metric.

7 Experimental Protocol

The experiment fixes the random seed, matrix dimension, operator family, final time, arithmetic precision, Krylov dimensions, fixed-step grids, adaptive tolerance, and reference procedure.

Table 2: Long-horizon benchmark contract.

Parameter	Value
Dimension N	3,000
Operator	s tridiag(1, -2 , 1)
Scale s	0.5, 2, 4
Final time T	20
Initial state	normalized pseudorandom vector
Seed	20260908
Krylov dimensions	12, 20, 28
Fixed step counts	10, 20, 40
Adaptive local tolerance	10^{-8}
Adaptive dimensions	20, 28
Reference	independent sparse exponential action
Arithmetic	IEEE double precision
Software	Python 3.13.5, NumPy 2.3.5, SciPy 1.17.0
Platform	Linux x86_64

The reference is computed independently as $e^{TA}v$ using a sparse matrix-exponential action routine. Candidate methods are compared by

$$\varepsilon_T = \frac{\|y_T - y_{\text{ref}}\|_2}{\|y_{\text{ref}}\|_2}. \quad (19)$$

Three timing observations are made for each fixed configuration after one warm-up execution. Adaptive configurations are timed over the complete accepted/rejected trajectory.

8 Results: Fixed-Step Krylov Propagation

For $s = 0.5$, the problem is numerically easy at the tested final time. Even $m = 12$ with ten steps reaches a relative error of 6.01×10^{-10} . Increasing the number of steps produces errors close to the

double-precision floor, but the additional sparse applications do not improve the scientific result materially.

For $s = 2$, the same $m = 12$ configuration requires more steps: ten steps produce a relative error of 1.03×10^{-4} , twenty steps produce 4.43×10^{-7} , and forty steps produce 6.24×10^{-10} . Increasing the Krylov dimension to $m = 20$ allows ten steps to reach 6.94×10^{-10} .

For $s = 4$, the coupling between time-step length and Krylov dimension becomes stronger. With $m = 12$, ten steps yield a relative error of 5.44×10^{-3} , while forty steps reduce the error to 4.84×10^{-7} . With $m = 20$, twenty steps achieve 7.13×10^{-10} , and $m = 28$ with ten steps reaches 1.59×10^{-10} .

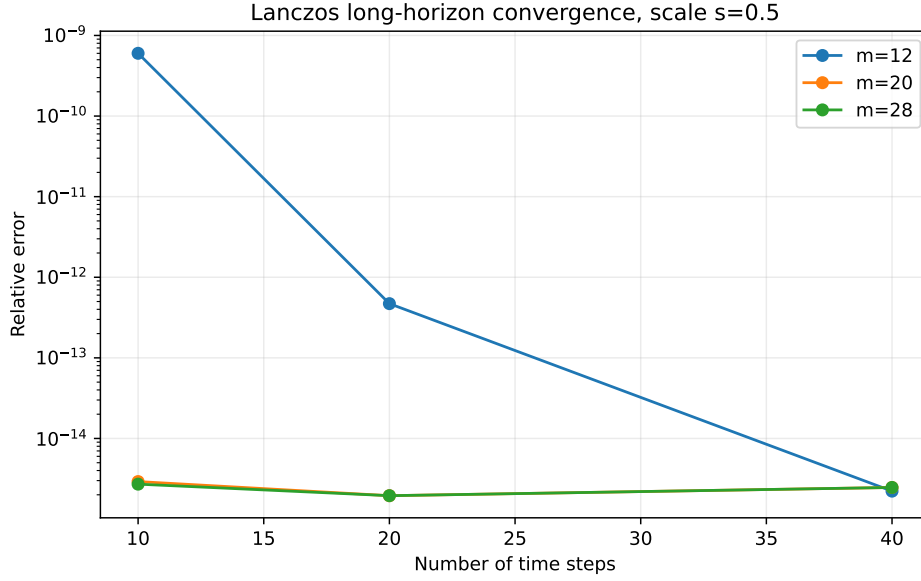


Figure 1: Fixed-step Lanczos convergence for $s = 0.5$. The smallest tested Krylov dimension already resolves the long-horizon propagation to near machine precision with a modest number of steps.

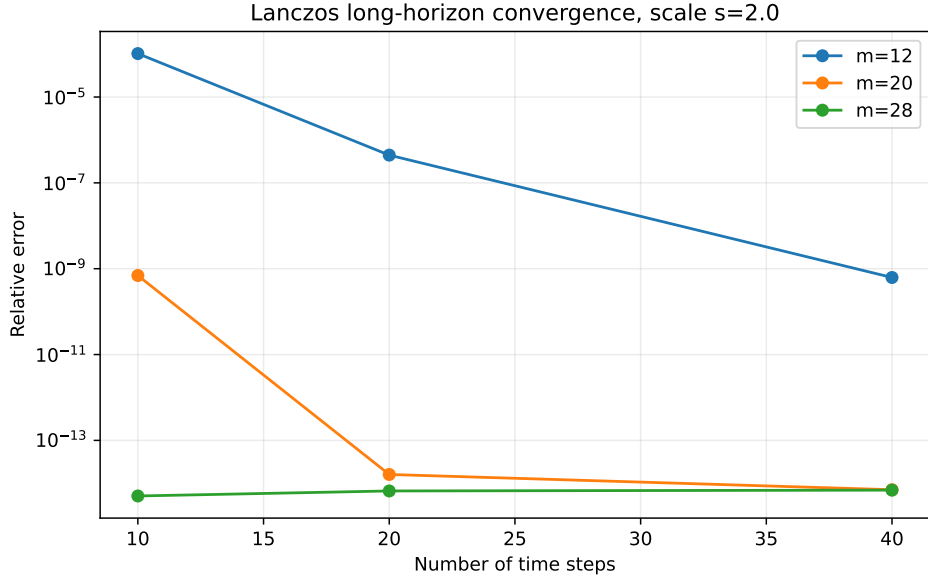


Figure 2: Fixed-step Lanczos convergence for $s = 2$. Increasing either the Krylov dimension or the number of steps improves the approximation, but their computational roles are different.

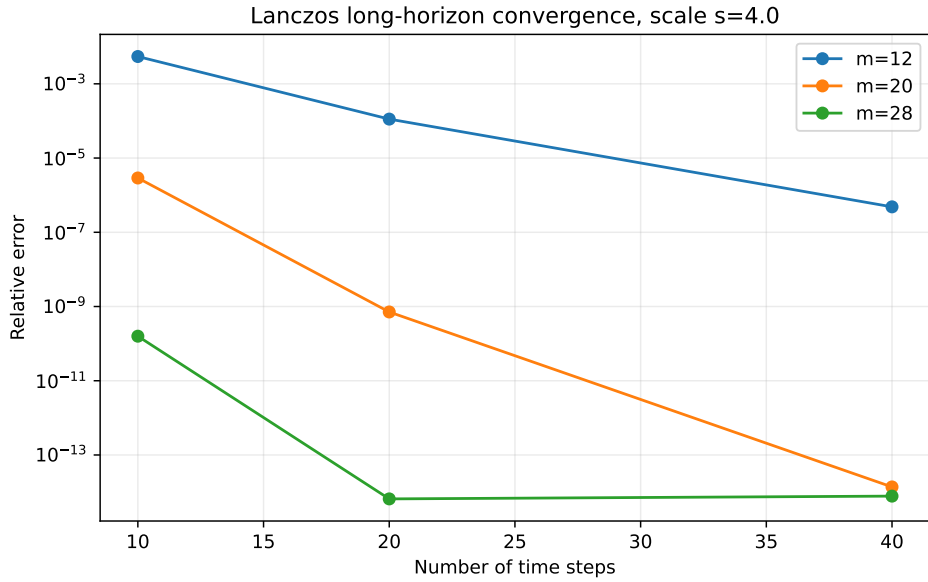


Figure 3: Fixed-step Lanczos convergence for $s = 4$. The stronger operator scale exposes the trade-off between larger Krylov spaces and shorter propagation steps.

9 Full Fixed-Step Benchmark Matrix

The complete measured fixed-step experiment is summarized below. The table is intentionally included because the same final error can arise from different combinations of Krylov dimension and time-step count. Reporting only the best configuration would conceal that algorithmic structure.

Table 3: Selected fixed-step results at $N = 3,000$.

s	Method	m	Steps	Error	Time (ms)
0.5	Lanczos	12	10	6.01×10^{-10}	6.92
0.5	Lanczos	12	20	4.71×10^{-13}	14.57
0.5	Lanczos	20	10	2.92×10^{-15}	29.95
0.5	Arnoldi	12	10	6.01×10^{-10}	12.77
0.5	Arnoldi	20	10	1.83×10^{-15}	19.16
2	Lanczos	12	10	1.03×10^{-4}	9.42
2	Lanczos	12	20	4.43×10^{-7}	14.92
2	Lanczos	12	40	6.24×10^{-10}	25.49
2	Lanczos	20	10	6.94×10^{-10}	48.26
2	Arnoldi	12	40	6.24×10^{-10}	36.55
2	Arnoldi	20	10	6.94×10^{-10}	19.01
4	Lanczos	12	10	5.44×10^{-3}	7.41
4	Lanczos	12	20	1.12×10^{-4}	16.11
4	Lanczos	12	40	4.84×10^{-7}	59.02
4	Lanczos	20	20	7.13×10^{-10}	30.18
4	Lanczos	28	10	1.59×10^{-10}	60.07
4	Arnoldi	20	20	7.13×10^{-10}	37.85
4	Arnoldi	28	10	1.59×10^{-10}	41.63

The table demonstrates why an accuracy target alone is insufficient for algorithm selection. For example, at $s = 4$, the target 10^{-9} can be reached by $m = 20$ with twenty steps or by $m = 28$ with ten steps. These configurations have different operator-application counts and different measured runtimes. The relevant optimization problem is therefore constrained by accuracy but not determined by accuracy alone.

10 Results: Lanczos versus Arnoldi

The symmetric benchmark permits a direct structural comparison between Lanczos and general Arnoldi. At the same m and number of steps, the two methods produce nearly identical approximation errors because both represent the same Krylov space. The computational difference is therefore attributable primarily to the projection construction and orthogonalization pattern rather than to a fundamentally different approximation space.

At $s = 2$, $m = 12$, and ten steps, Lanczos and Arnoldi produce relative errors of 1.0275×10^{-4} and 1.0275×10^{-4} , respectively. At $s = 4$, $m = 20$, and twenty steps, the corresponding errors are 7.1284×10^{-10} and 7.1284×10^{-10} . These matched errors provide a useful internal check: the observed numerical difference is not caused by different mathematical target spaces.

The timing results are more variable because the benchmark is small enough that cache behavior, Python overhead, and dense projected-exponential cost are visible. Consequently, the paper does not infer a universal speedup factor from this experiment. The scientifically stable conclusion is

instead structural: when symmetry is known, Lanczos reduces the projected matrix to tridiagonal form and removes the need to construct a full Hessenberg projection.

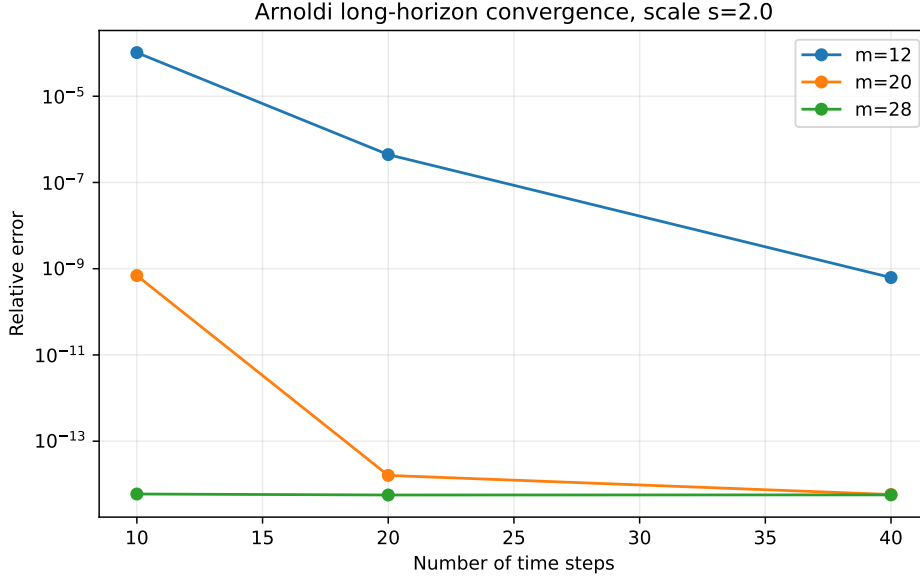


Figure 4: Fixed-step Arnoldi convergence for $s = 2$. The error trajectory closely matches the corresponding Krylov-space behavior of Lanczos because both span the same Krylov subspace at equal dimension.

11 Results: Adaptive Step Control

The adaptive strategy changes the computational balance. For $s = 0.5$ and $m = 20$, the controller accepts four steps and rejects none, producing a final relative error of 2.43×10^{-15} with 240 sparse operator applications. For $m = 28$, it also accepts four steps, with 336 operator applications and a final error of 2.37×10^{-15} .

For $s = 2$, $m = 20$ requires five accepted steps, no rejected steps, 300 operator applications, and a final relative error of 1.07×10^{-13} . With $m = 28$, four accepted steps suffice, giving 336 operator applications and 6.36×10^{-15} error.

The most informative regime is $s = 4$. With $m = 20$, the controller accepts seven steps and rejects three. The resulting trajectory uses 420 sparse operator applications and reaches a final relative error of 7.17×10^{-14} . Increasing the Krylov dimension to $m = 28$ eliminates the rejections: four accepted steps are sufficient, with 336 operator applications and a final error of 2.43×10^{-14} .

This result illustrates a concrete algorithmic trade-off. A larger Krylov space can increase the cost per step while reducing the number of steps and failed trials. A smaller space can reduce the projected dimension but may force the controller to take shorter steps or reject more trials. The adaptive method therefore exposes a two-dimensional tuning problem rather than a single parameter.

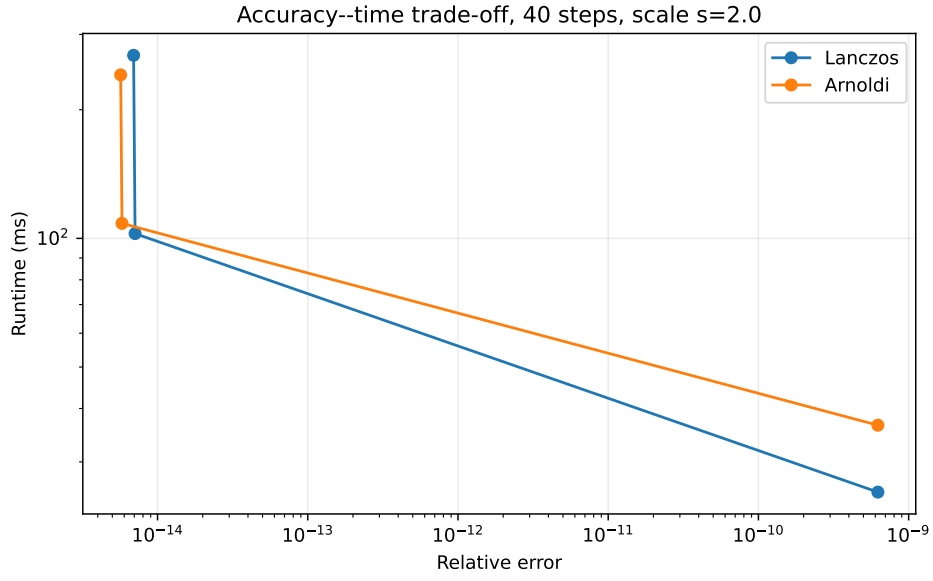


Figure 5: Observed runtime–accuracy trade-off at forty fixed steps for $s = 2$. The plot is an empirical comparison for the specified implementation and hardware.

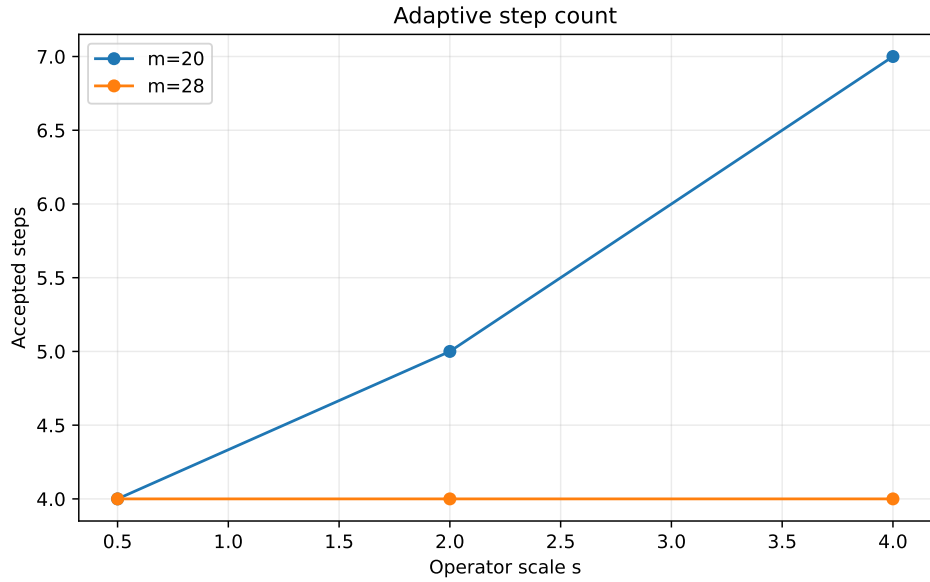


Figure 6: Number of accepted adaptive steps as a function of operator scale. The controller changes the global step count according to the observed local difficulty.

Table 4: Measured adaptive Lanczos results.

s	m	Accepted	Rejected	Time (ms)	Relative error	Operator applications
0.5	20	4	0	16.59	2.43×10^{-15}	240
0.5	28	4	0	61.99	2.37×10^{-15}	336
2	20	5	0	20.61	1.07×10^{-13}	300
2	28	4	0	60.79	6.36×10^{-15}	336
4	20	7	3	79.76	7.17×10^{-14}	420
4	28	4	0	29.11	2.43×10^{-14}	336

12 Accuracy–Work Interpretation

The experiments support a useful interpretation of adaptive exponential integration. The local discrepancy is not merely an error number; it is a mechanism for deciding whether computational effort should be spent on a larger Krylov space or on additional time resolution.

At $s = 4$, the $m = 20$ controller demonstrates the cost of an undersized projection space. It rejects three trials and ultimately performs 420 operator applications. The $m = 28$ controller uses 336 applications and no rejected steps. In this particular implementation, the larger projected problem is therefore more efficient despite the greater per-step projection dimension.

The result should not be generalized into the statement that larger Krylov spaces are always better. At $s = 0.5$, $m = 20$ and $m = 28$ both use four accepted steps, so the additional basis vectors do not buy additional global step reduction. The smaller space is consequently preferable in operator-application count and observed runtime.

A practical selection rule follows directly from the experiment: begin with a moderate Krylov dimension, monitor the step-doubling discrepancy, and enlarge the dimension when repeated rejections indicate that time-step reduction is becoming more expensive than projection enlargement. This is a policy derived from the measured benchmark, not a universal theorem.

13 Verification of the Adaptive Indicator

The adaptive indicator is intentionally treated as a controller rather than as a certified global error bound. The final error is independently measured against the reference solution. This distinction is especially important because local discrepancies can accumulate over many accepted steps.

For every tested adaptive configuration, the final reference error is substantially below the local tolerance 10^{-8} . The largest final error in the adaptive table is 1.07×10^{-13} , observed at $s = 2$, $m = 20$. The result confirms that the controller is conservative for the tested negative-semidefinite family.

The experiment does not claim that the same tolerance will produce the same global error for non-normal operators, time-dependent operators, nonlinear Jacobians, or operators with strong transient growth. Those cases require a separate validation study.

14 Connection to Exponential Integrators

The numerical step in equation (11) is an exponential integrator for the homogeneous linear system. For a semilinear system,

$$y'(t) = Ay(t) + g(t, y(t)), \quad (20)$$

exponential integrators also require the related φ -functions. Krylov algorithms for these functions have been developed specifically for large-scale matrix-free integration, including adaptive procedures that vary both time-step size and Krylov dimension. [3, 4]

The present study deliberately stops at the homogeneous linear layer. This gives a controlled environment in which the effect of symmetry, step size, and Krylov dimension can be measured without confounding nonlinear Jacobian updates or nonlinear residual evaluations. A natural extension is to introduce a QFLPN-derived forcing term and evaluate the resulting φ -function actions.

15 Limitations

The benchmark uses a symmetric tridiagonal operator family. This is scientifically useful because it isolates the Lanczos advantage, but it does not represent every sparse QFLPN operator. Non-symmetric and non-normal operators may require Arnoldi rather than Lanczos, and the convergence behavior can be governed by the numerical range or pseudospectrum rather than only by eigenvalues. [2]

The state dimension is $N = 3,000$. The experiment demonstrates long-horizon behavior and adaptive control, but it is not a large-scale hardware benchmark. The measured runtimes should therefore not be interpreted as performance predictions for millions of dimensions.

The adaptive controller uses a single local tolerance and a simple step-doubling policy. More sophisticated algorithms can use residual estimates, variable Krylov dimensions, incomplete orthogonalization, or direct control of φ -function combinations. The present paper establishes a transparent baseline against which such extensions can be compared.

16 Reproducibility and Artifact Boundary

The numerical artifact contains the complete fixed-step benchmark, adaptive benchmark, summary table, random seed, operator definition, dimensions, time horizon, Krylov dimensions, tolerance, software versions, and reference definition. The generated figures are derived directly from those CSV records.

No result in this paper is inferred from the single-step CSR/SpMV timing experiment. The measured quantities here are generated specifically for the long-horizon integration problem. Likewise, no result is transferred from a small four-qubit operator construction into this sparse numerical experiment. This separation is intentional: each study has a distinct mathematical object, experimental question, and evidence set.

17 Conclusion

This paper developed and experimentally evaluated an adaptive Krylov–Lanczos exponential integrator for long-horizon sparse state evolution. The central finding is that time-step length and Krylov dimension should be treated as coupled computational controls. On the tested symmetric negative-semidefinite family, adaptive step-doubling achieved final errors far below the local tolerance while reducing the number of global propagation stages. For the strongest tested scale, $s = 4$, increasing the Krylov dimension from 20 to 28 reduced rejected work from three trials to zero and reduced operator applications from 420 to 336.

The result is not a universal ranking of Lanczos over Arnoldi or of adaptive over fixed-step propa-

gation. Its value is the reproducible separation of three mechanisms: sparse operator application, projected matrix-function approximation, and global time-step control. This decomposition provides a technically clean basis for extending the study to non-symmetric QFLPN operators, nonlinear transition layers, and φ -function exponential integrators.

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